1. Simplify the expression below into the form $a + bi$. Then, choose the intervals that a and b belong to.

$$\frac{9 + 44i}{-5 + 8i}$$

   A. $3.45 - 3.28i$

   B. $-1.80 + 5.50i$

   C. $-4.46 - 1.66i$

   D. $3.45 - 292.00i$

   E. $307.00 - 3.28i$

---

2. Simplify the expression below into the form $a + bi$. Then, choose the intervals that a and b belong to.

$$(-5 - 3i)(-10 + 4i)$$

   A. $62.00 + 10.00i$

   B. $38.00 - 50.00i$

   C. $38.00 + 50.00i$

   D. $62.00 - 10.00i$

   E. $50.00 - 12.00i$

---

3. Simplify the expression below and choose the interval the simplification is contained within.

$$6 - 20^2 + 12 \div 16 * 19 \div 17$$

   A. -393.162

   B. 406.838

   C. -393.998

   D. 406.002

E. $Noneoftheabove$

---

4. Choose the **smallest** set of Real numbers that the number below belongs to.

$$\sqrt{\frac{-1666}{14}}$$

A. Whole

B. Integer

C. Rational

D. Irrational

E. Not a Real Number

---

5. Choose the **smallest** set of Complex numbers that the number below belongs to.

$$\frac{\sqrt{156}}{9} + \sqrt{-9}i$$

A. Rational

B. Irrational

C. Nonreal Complex

D. Pure Imaginary

E. Not a Complex Number

---

6. Solve the equation below. Then, choose the interval that contains the solution.

$$-5(-18x - 11) = -14(2x + 19)$$

A. -2.72

B. -1.788

C. 1.788

D. 3.403

E. There are no Real solutions.

---

7. Solve the linear equation below. Then, choose the interval that contains the solution.

$$\frac{-3x+9}{8} - \frac{-3x-9}{5} = \frac{-4x+5}{6}$$

A. -2.346

B. 1.692

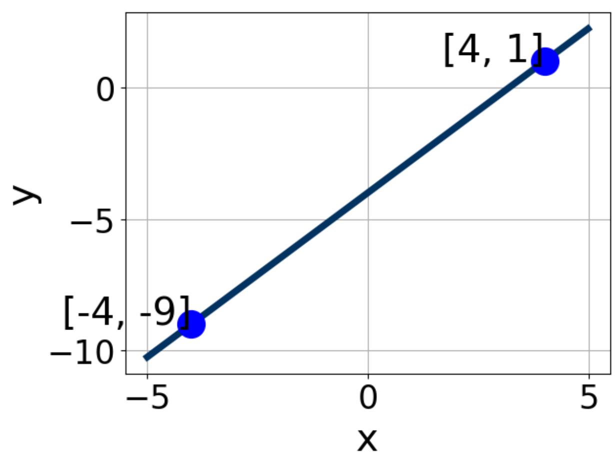
C. -14.579

D. -0.523

E. There are no Real solutions.

---

8. Write the equation of the line in the graph below in Standard Form $Ax + By = C$. Then, choose the intervals that contain $A, B,$ and C.



A. $5x - 4y = 16$

B. $-5x + 4y = -16$

C. $5x + 4y = -16$

D. $-1.25x + 1y = -4.0$

E. $-1.25x - 1y = 4.0$

9. First, find the equation of the line containing the two points below. Then, write the equation in the form $y = mx+b$ and choose the intervals that contain m and b.

$$(-9, -10) \text{ and } (-7, -7)$$

A. $y = 1.5x + 3.5$

B. $y = -1.5x - 17.5$

C. $y = 1.5x - 3.5$

D. $y = 1.5x - 1$

E. $y = 1.5x$

---

10. Find the equation of the line described below. Write the linear equation in the form $y = mx + b$ and choose the intervals that contain m and b.

Parallel to $4x + 7y = 5$ and passing through the point $(-9, 3)$.

A. $y = -0.57x - 2.14$

B. $y = 0.57x + 8.14$

C. $y = -0.57x + 2.14$

D. $y = -0.57x + 12.00$

E. $y = -1.75x - 2.14$

---

11. Using an interval or intervals, choose the option that describes all the x-values within or including a distance of the given values.

No more than 10 units from the number 7.

A. $[3, 17]$

B. $(3, 17)$

C. $(-\infty, 3) \cup (17, \infty)$

D. $(-\infty, 3] \cup [17, \infty)$

E. None of the above

---

12. Solve the linear inequality below. Then, choose the constant and interval combination that describes the solution set.

$$-5 + 6x \leq \frac{27x + 7}{4} < 6 + 5x$$

A. $(-9.0, 2.4285714285714284]$

B. $[-9.0, 2.4285714285714284)$

C. $(-\infty, -9.0) \cup [2.4285714285714284, \infty)$

D. $(-\infty, -9.0] \cup (2.4285714285714284, \infty)$

E. None of the above.

---

13. Solve the linear inequality below. Then, choose the constant and interval combination that describes the solution set.

$$-4 + 9x > 10x \text{ or } 6 + 4x < 7x$$

A. $(-\infty, -4.0)$ or $(2.0, \infty)$

B. $(-\infty, -2.0)$ or $(4.0, \infty)$

C. $(-\infty, -4.0]$ or $[2.0, \infty)$

D. $(-\infty, -2.0]$ or $[4.0, \infty)$

E. $(-\infty, \infty)$

---

14. Solve the linear inequality below. Then, choose the constant and interval combination that describes the solution set.

$$-10x - 3 \leq 3x + 5$$

A. $[-0.615, \infty)$

B. $(-\infty, -0.615]$

C. $[0.615, \infty)$

D. $(-\infty, 0.615]$

E. None of the above

---

15. Solve the linear inequality below. Then, choose the constant and interval combination that describes the solution set.

$$\frac{-3}{9} + \frac{3}{4}x < \frac{7}{6}x - \frac{5}{2}$$

A. $(5.2, \infty)$

B. $(-\infty, 5.2)$

C. $(-5.2, \infty)$

D. $(-\infty, -5.2)$

E. None of the above

---

16. Solve the quadratic equation below. Then, choose the intervals that the solutions x_1 and x_2 belong to, with $x_1 \leq x_2$.

$$6x^2 - 35x + 36$$

A. $x_1 = 1.333$ and $x_2 = 4.500$

B. $x_1 = 8.000$ and $x_2 = 27.000$

C. $x_1 = 1.500$ and $x_2 = 4.000$

D. $x_1 = 2.250$ and $x_2 = 2.667$

E. $x_1 = 0.444$ and $x_2 = 13.500$

---

17. Factor the quadratic below. Then, choose the intervals that contain the constants in the form $(ax + b)(cx + d); b \leq d$.

$$100x^2 + 140x + 49$$

A. $(10x + 7)(10x + 7)$

B. $(x + 70)(x + 70)$

C. $(5x + 7)(20x + 7)$

D. $(50x + 7)(2x + 7)$

E. None of the above.

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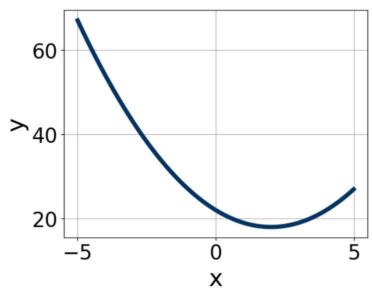
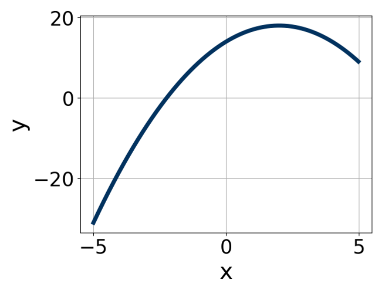
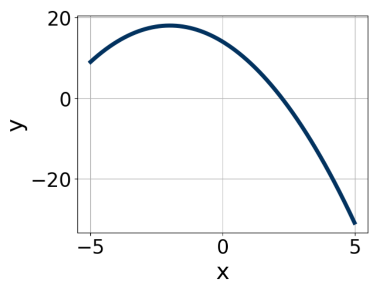
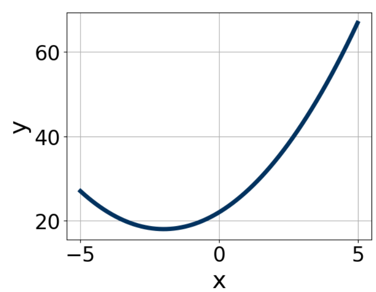
18. Choose the graph of the equation below.

$$f(x) = (x - 2)^2 + 18$$

A. 

C. 

B. 

D. 

E. None of the above.

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19. Solve the quadratic equation below. Then, choose the intervals that the solutions belong to, with $x_1 \leq x_2$ (if they exist).

$$16x^2 - 15x + 3$$

A. $x_1 = 0.289$ and $x_2 = 0.648$

B. $x_1 = 0.203$ and $x_2 = 14.797$

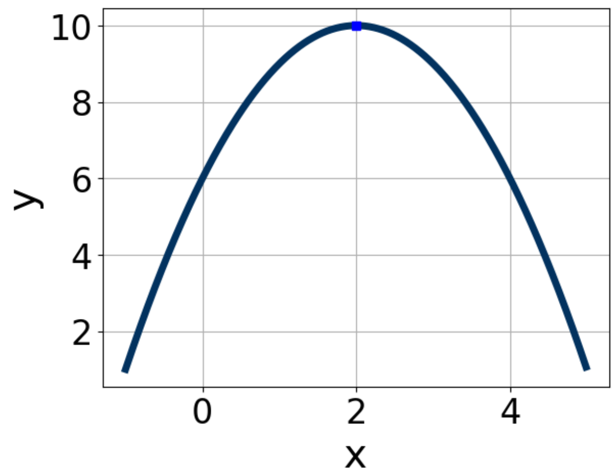
C. $x_1 = -0.648$ and $x_2 = -0.289$

D. $x_1 = -5.276$ and $x_2 = 6.213$

E. There are no Real solutions

---

20. Write the equation of the graph presented below in the form $f(x) = ax^2 + bx + c$, assuming $a = 1$ or $a = -1$. Then, choose the intervals that $a, b,$ and c belong to.



A. $-x^2 + 4x + 6$

B. $-x^2 - 4x + 6$

C. $x^2 + 4x + 14$

D. $-x^2 - 4x - 14$

E. $x^2 - 4x + 14$
